EX.NO:3 INHERITANCE PROGRAM FOR AN EMPLOYEE CLASS

PROGRAM:

```
import java.util.Scanner;

class Vehicle {
    String vehicleNumber;
    String model;
    String manufacturer;
    double price;

    Vehicle(String vehicleNumber, String model, String manufacturer, double price) {
        this.vehicleNumber = vehicleNumber;
        this.model = model;
        this.manufacturer = manufacturer;
        this.price = price;
    }

    void display() {
        System.out.println("Vehicle Number : " + vehicleNumber);
        System.out.println("Model          : " + model);
        System.out.println("Manufacturer   : " + manufacturer);
        System.out.println("Price          : " + price);
    }
}

class Car extends Vehicle {
    Car(String v, String m, String man, double p) {
        super(v, m, man, p);
    }

    void bill() {
        double tax = price * 0.05;
        double insurance = 10000;
        double total = price + tax + insurance;

        System.out.println("\n***** CAR BILL *****");
        display();
        System.out.println("Road Tax   : " + tax);
        System.out.println("Insurance  : " + insurance);
        System.out.println("Total Cost : " + total);
    }
}

class Bike extends Vehicle {
    Bike(String v, String m, String man, double p) {
        super(v, m, man, p);
    }

    void bill() {
        double tax = price * 0.03;
        double insurance = 3000;
        double total = price + tax + insurance;

        System.out.println("\n***** BIKE BILL *****");
        display();
        System.out.println("Road Tax   : " + tax);
        System.out.println("Insurance  : " + insurance);
```

```
        System.out.println("Total Cost : " + total);
    }
}

class Truck extends Vehicle {
    Truck(String v, String m, String man, double p) {
        super(v, m, man, p);
    }

    void bill() {
        double tax = price * 0.10;
        double insurance = 15000;
        double total = price + tax + insurance;

        System.out.println("\n***** TRUCK BILL *****");
        display();
        System.out.println("Road Tax   : " + tax);
        System.out.println("Insurance  : " + insurance);
        System.out.println("Total Cost : " + total);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Vehicle Number: ");
        String number = sc.nextLine();

        System.out.print("Enter Model: ");
        String model = sc.nextLine();

        System.out.print("Enter Manufacturer: ");
        String manufacturer = sc.nextLine();

        System.out.print("Enter Vehicle Price: ");
        double price = sc.nextDouble();

        System.out.println("\n1. Car");
        System.out.println("2. Bike");
        System.out.println("3. Truck");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {
            case 1:
                Car c = new Car(number, model, manufacturer, price);
                c.bill();
                break;

            case 2:
                Bike b = new Bike(number, model, manufacturer, price);
                b.bill();
                break;

            case 3:
                Truck t = new Truck(number, model, manufacturer, price);
                t.bill();
                break;

            default:
```

```
                System.out.println("Invalid Choice");
            }

            sc.close();
        }
    }
```

OUTPUT:

Sample Run 1 (Car)

```
Enter Vehicle Number: TN32AB1234
Enter Model: Swift
Enter Manufacturer: Maruti
Enter Vehicle Price: 800000

1. Car
2. Bike
3. Truck
Enter Choice: 1

***** CAR BILL *****
Vehicle Number : TN32AB1234
Model          : Swift
Manufacturer   : Maruti
Price          : 800000.0
Road Tax   : 40000.0
Insurance  : 10000.0
Total Cost : 850000.0
```

Sample Run 2 (Bike)

```
Enter Vehicle Number: TN09XY4321
Enter Model: Splendor
Enter Manufacturer: Hero
Enter Vehicle Price: 90000

1. Car
2. Bike
3. Truck
Enter Choice: 2

***** BIKE BILL *****
Vehicle Number : TN09XY4321
Model          : Splendor
Manufacturer   : Hero
Price          : 90000.0
Road Tax   : 2700.0
Insurance  : 3000.0
Total Cost : 95700.0
```

Sample Run 3 (Truck)

```
Enter Vehicle Number: TN01TK9999
Enter Model: Tata 407
Enter Manufacturer: Tata
Enter Vehicle Price: 1200000

1. Car
2. Bike
3. Truck
Enter Choice: 3

***** TRUCK BILL *****
Vehicle Number : TN01TK9999
Model          : Tata 407
Manufacturer   : Tata
Price          : 1200000.0
Road Tax   : 120000.0
```

```
Insurance  : 15000.0
Total Cost : 1335000.0
```